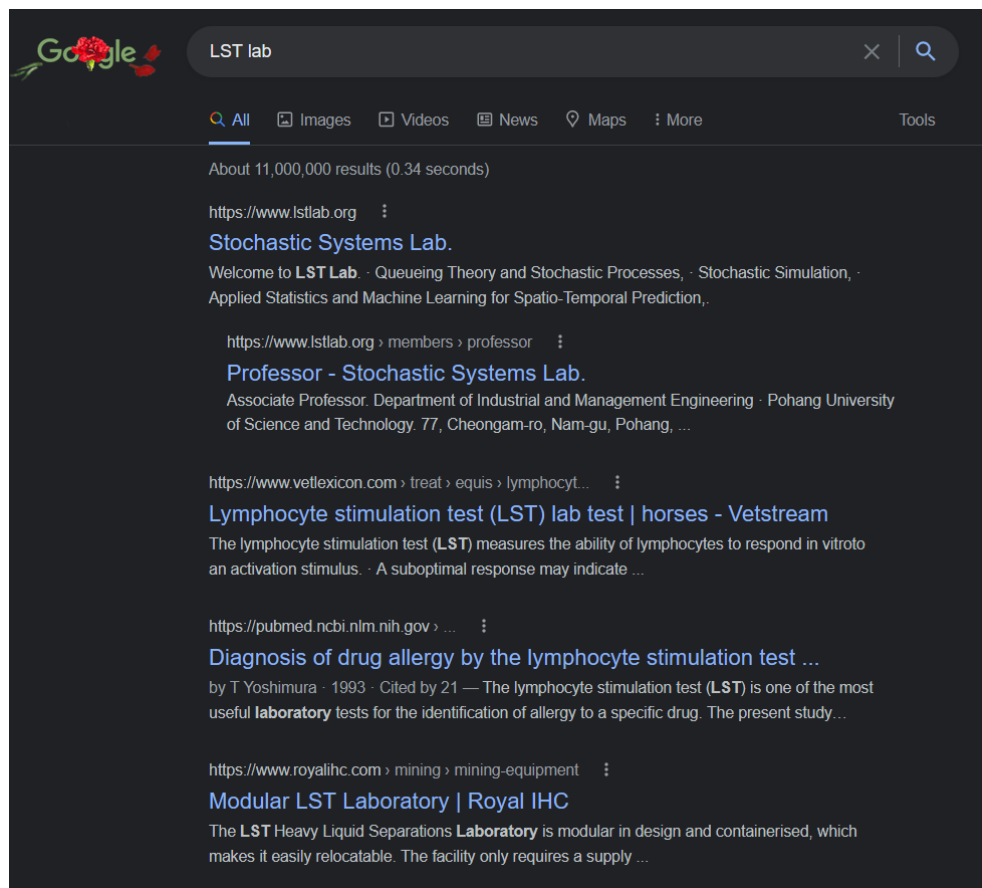


Google PageRank

Markov Chains

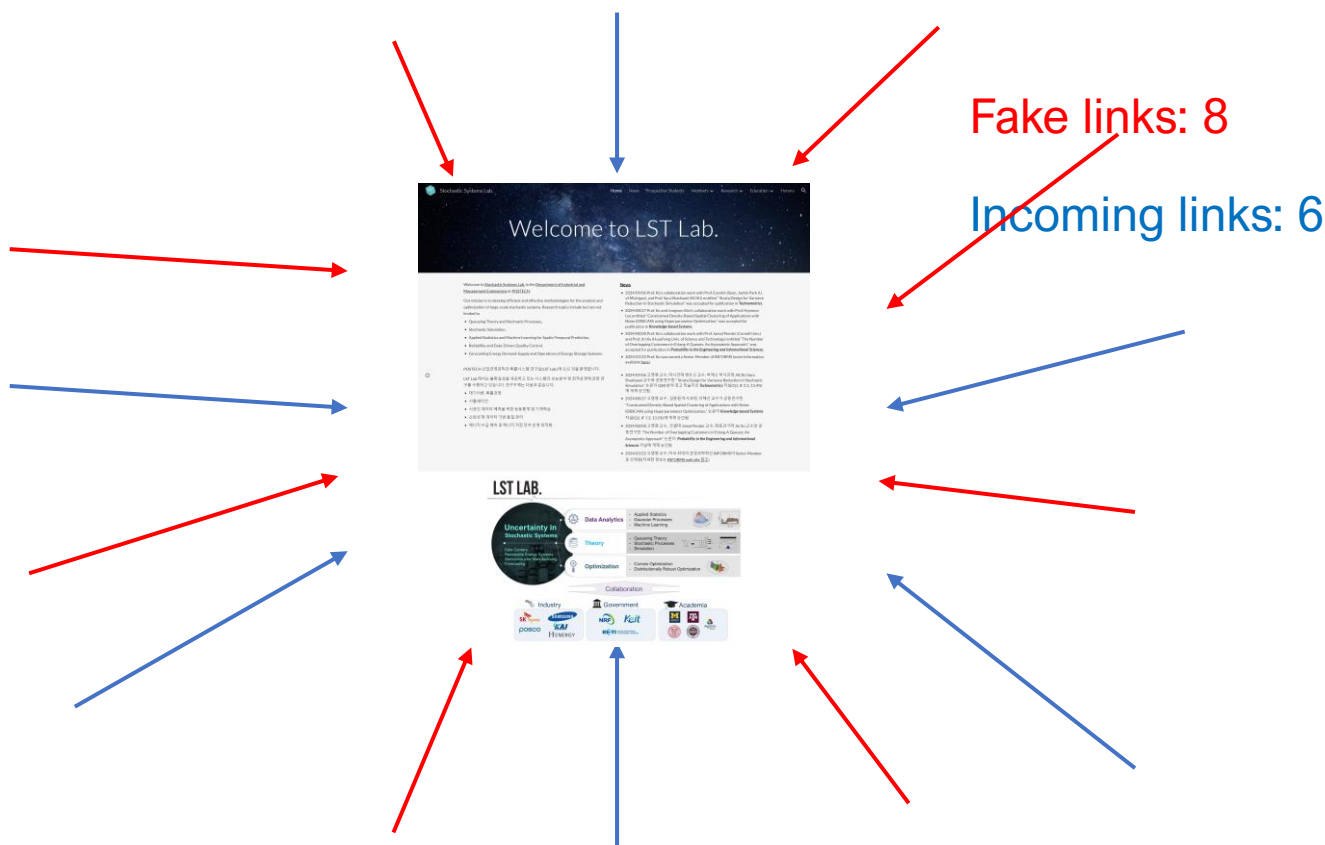
Problem

- How to show web pages to users?



Ranking web pages?

- First idea: citation counting



Any Idea?

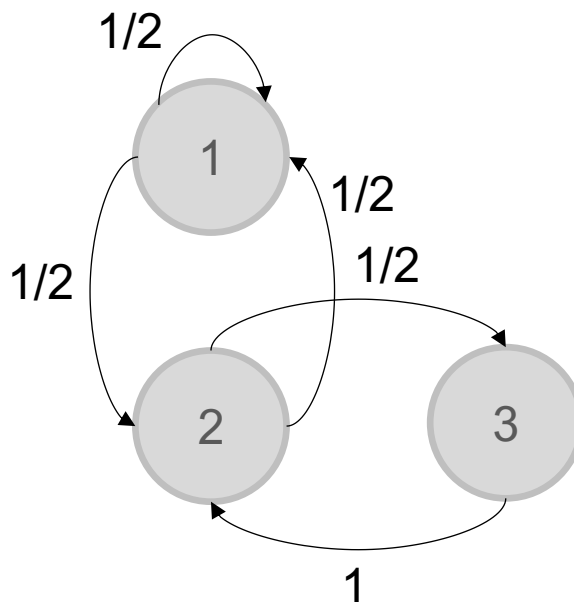
- Determine rank recursively using
Markov Chain!!

$$\pi = \pi P$$

$$\pi_j = \sum_{i=1}^n \pi_i P_{ij}$$

PageRank Algorithm

- Rule: If Page i has k outgoing links, each link has probability $1/k$.
- Construct P matrix.
- Solve $\pi = \pi P$.



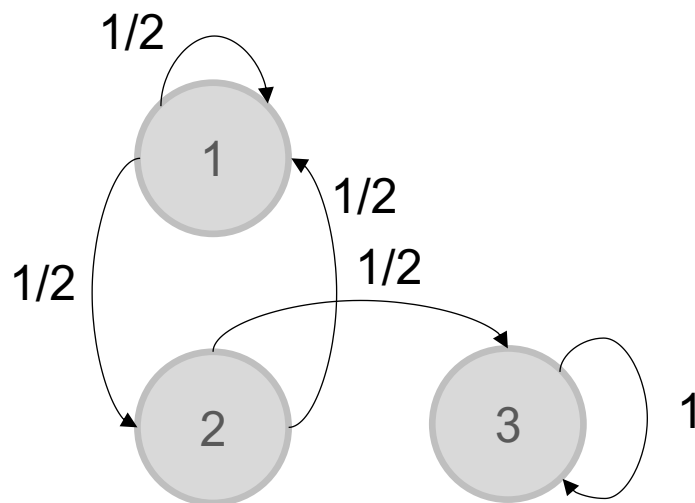
$$\pi_1 = 0.4$$

$$\pi_2 = 0.4$$

$$\pi_3 = 0.2$$

Dead End or Spider Trap

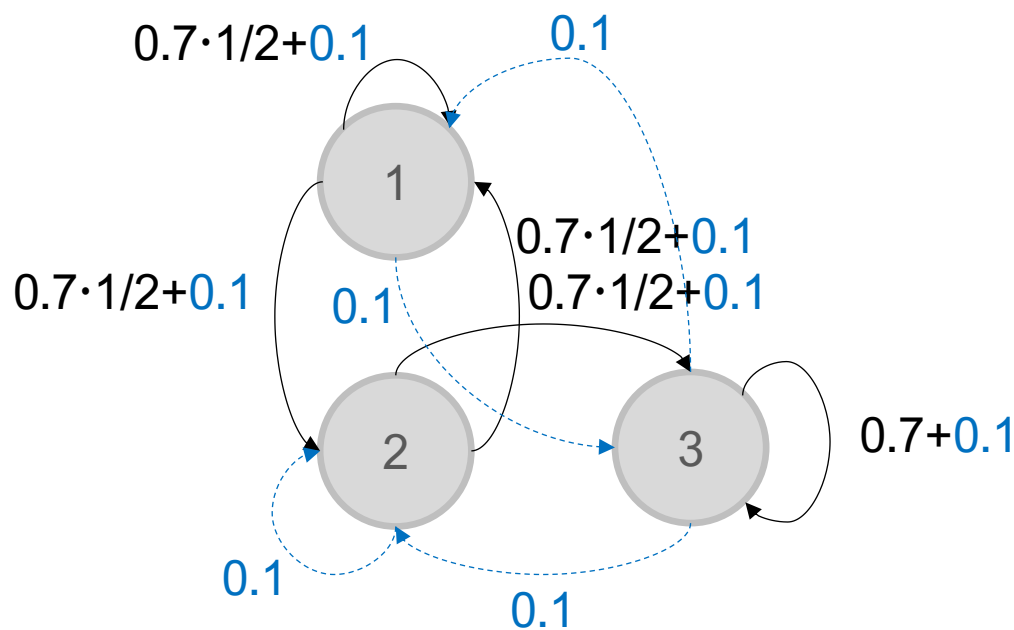
- Violation of ergodicity



Taxation

- Collect *tax* and distribute *equally* to every node

30% Tax



$$\pi_1 = 0.26$$

$$\pi_2 = 0.19$$

$$\pi_3 = 0.55$$